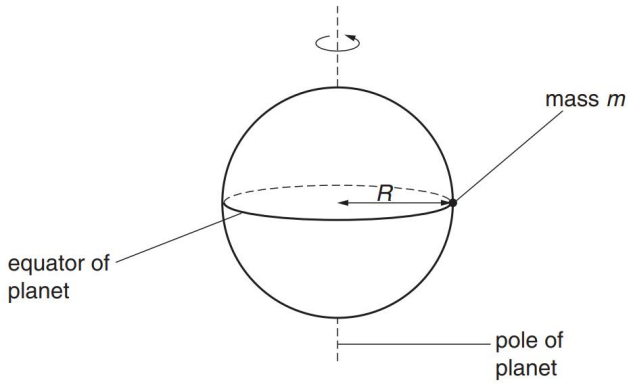
13.1.2 Exercises - Newton's law of gravitation 2

1. A spherical planet has mass *M* and radius *R*.

The planet may be assumed to be isolated in space and to have its mass concentrated at its centre.

The planet spins on its axis with angular speed ω, as illustrated in Fig.



A small object of mass m rests on the equator of the planet. The surface of the planet exerts a normal reaction force on the mass.

a. State formulae, in terms of *M*, *m*, *R* and *ω*, for

i. the gravitational force between the planet and the object,

.....[1]

ii. the centripetal force required for circular motion of the small mass,

.....[1]

iii. the normal reaction exerted by the planet on the mass.

.....[1]

b.

i. Explain why the normal reaction on the mass will have different values at the equator and at the poles.

.....

.....

.....[2]

ii. The radius of the planet is 6.4×10^6 m. It completes one revolution in 8.6×10^4 s.

Calculate the magnitude of the centripetal acceleration at

1. the equator,

acceleration = ... ms^{-2} [2]

2. one of the poles.

acceleration = ... ms^{-2} [1]

c. Suggest two factors that could, in the case of a real planet. cause variations in the acceleration of free fall at its surface.

1. .....

.....

2. .....

.....[2]

2.

a. A moon is in a circular orbit of radius r about a planet. The angular speed of the moon in its orbit is ω. The planet and its moon may be considered to be point masses that are isolated in space. Show that rand ω are related by the expression

$$r^3\omega^2 = constant$$

Explain your working. [3]

b. Phobos and Deimos are moons that are in circular orbits about the planet Mars.
Data for Phobos and Deimos are shown in Fig.

moon	radius of orbit / m	period of rotation about Mars / hours
Phobos Deimos	9.39×10^6 1.99×10^7	7.65

i. Use data from Fig. to determine

1. the mass of Mars,

mass = ... kg [3]

2. the period of Deimos in its orbit about Mars.

period = ... hours [3]

ii. The period of rotation of Mars about its axis is 24.6 hours.

Deimos is in an equatorial orbit, orbiting in the same direction as the spin of Mars about its axis. Use your answer in **(i)** to comment on the orbit of Deimos.

.....[1]

3.

a. Explain what is meant by a *geostationary orbit*.

.....

.....

.....[3]

b. A satellite of mass *m* is in a circular orbit about a planet.

The mass *M* of the planet may be considered to be concentrated at its centre.

Show that the radius *R* of the orbit of the satellite is given by the expression

$$R^3 = \frac{GMT^2}{4\pi^2}$$

where *T* is the period of the orbit of the satellite and *G* is the gravitational constant.

Explain your working. [4]

c. The Earth has mass 6.0×10^{24} kg. Use the expression given in **(b)** to determine the radius of the geostationary orbit about the Earth.

radius = ... m [3]

4.

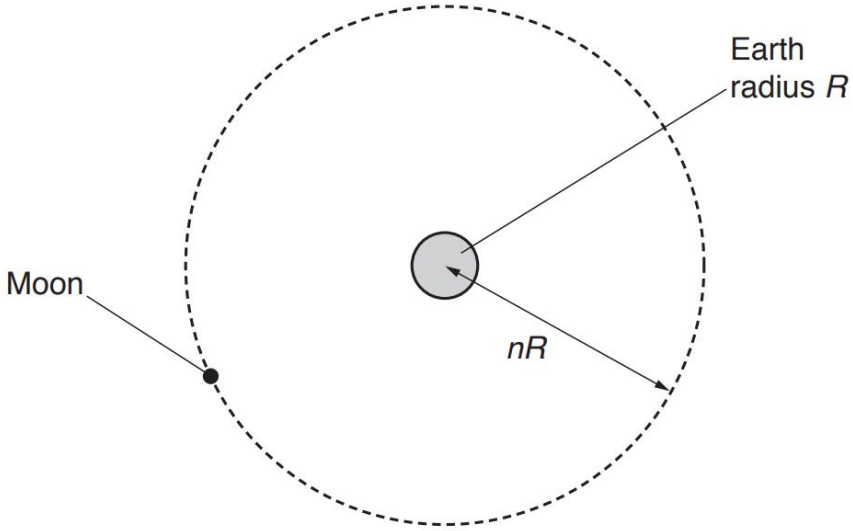
a. Explain how a satellite may be in a circular orbit around a planet.

.....

.....

.....[2]

b. The Earth and the Moon may be considered to be uniform spheres that are isolated in space. The Earth has radius *R* and mean density *ρ*. The Moon, mass *m*, is in a circular orbit about the Earth with radius *nR*, as illustrated in Fig.



The Moon makes one complete orbit of the Earth in time *T*.
Express the mean density *ρ* of the Earth [4]

c. The radius *R* of the Earth is 6.38×10^3 km and the distance between the centre of the Earth and the centre of the Moon is 3.84×10^5 km.
The period *T* of the orbit of the Moon about the Earth is 27.3 days.
Use the expression in **(b)** to calculate *ρ*.

ρ = ... $kg\ m^{-3}$ [3]